

Warm-starting unblocked QFT from a trained blocked basis

Question. The headline experiment trains the unblocked `qft` basis at $m = n = 8$ on DIV2K-8q to 31.29 dB at $\rho = 0.20$, while `blocked_8` (a `BlockedBasis(QFTBasis(3, 3), 5, 5)` whose operator is $\text{QFT}(3,3) \otimes \text{QFT}(3,3)$ replicated across a 32×32 grid of 8×8 blocks) reaches 32.26 dB — a 0.97 dB gap from a strictly **less** parametrically expressive family. $\text{QFT}(8, 8)$ has 288 trainable tensor entries; $\text{QFT}(3, 3)$ has 48. The blocked basis is a special low-dimensional submanifold of $\text{QFT}(8, 8)$, and yet random-init Adam on the larger family ends up worse. Is the blocked optimum reachable from inside the $\text{QFT}(8, 8)$ family, or is it structurally outside?

Construction. The trained blocked solution is a specific point in the $\text{QFT}(8, 8)$ parameter space — by construction. We embed it explicitly:

Take the trained 48 tensor entries of `BlockedBasis(QFTBasis(3,3), block_log_m=5, block_log_n=5).inner`. Place them in the $\text{QFT}(8, 8)$ circuit at the qubit positions corresponding to the inner block. Pin every other gate to identity ($H \rightarrow I_2$, controlled-phase $\rightarrow$ phase 0, equivalently $[[1, 1], [1, 1]]$). The resulting `QFTBasis(8, 8)` has all 288 parameters present and trainable, but its initial operator equals the trained blocked operator **bit-exactly**: $\max_{i,j} |T_{\text{warm}} x - T_{\text{blocked}} x|_{ij} = 0$ on a random complex 256×256 test input.

After this warm-start, we train the full $\text{QFT}(8, 8)$ under the same preset that produced the headline `qft` and `blocked_8` numbers (1008 steps, batch 50, validation split 0.15, Adam with cosine LR warmup, no early stopping).

Result (QuickDraw, $m = n = 5$). The same construction with inner = $\text{QFT}(3, 3)$ embedded in a 4×4 block grid (so $\text{QFT}(5, 5)$, 30 tensors total). After 1008 steps:

basis	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
qft (random init)	16.72	19.58	22.05	24.35
blocked (warm-start source)	18.12	22.40	26.20	30.05
qft_warmstart_blocked	18.13	22.41	26.22	30.06

The warm-started run **reproduces the trained blocked PSNR to within 0.02 dB at every keep-ratio**. Random-init `qft` lags by 5.7 dB at $\rho = 0.20$. So on QuickDraw the trained blocked optimum is reachable from inside the $\text{QFT}(5, 5)$ parameter family — what was missing from the headline `qft` run was **initialization**, not parametric expressivity.

Result (DIV2K-8q, $m = n = 8$). The headline gap is 0.97 dB at $\rho = 0.20$ (`qft` 31.29 vs `blocked_8` 32.26). After warm-start and full re-training:

basis	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
qft (random init)	25.09	27.57	29.53	31.29
blocked_8 (warm-start source)	25.18	28.09	30.30	32.26
qft_warmstart_blocked_8	25.18	28.09	30.30	32.26

The warm-started run reproduces **blocked_8 to two decimal places at every keep-ratio**. Random-init `qft` lags by 0.52, 0.77, and 0.97 dB at $\rho = 0.10$, 0.15, and 0.20 respectively. So on DIV2K too the trained blocked optimum sits inside the $\text{QFT}(8, 8)$ family and is reached by Adam from this warm-start init.

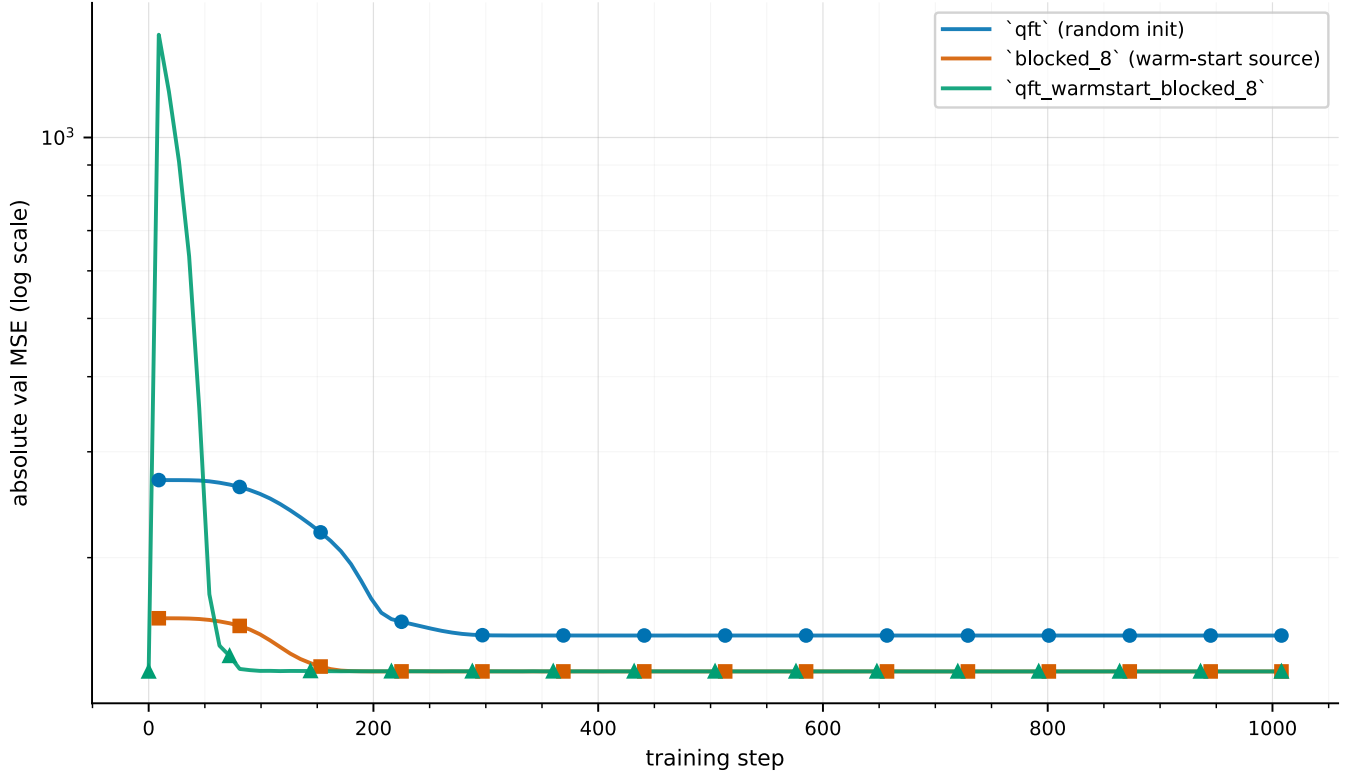


Figure 1: DIV2K-8q. Validation MSE per training step (log y). Blue: `qft` from random init. Orange: `blocked_8` headline run (warm-start source). Green: `qft_warmstart_blocked_8`. **The green curve’s step-0 marker is plotted at exactly the trained blocked floor** — by the bit-exact identity, the warm-start’s val MSE at step 0 equals `blocked_8`’s converged val MSE (sampled training points are recorded only after each completed epoch, so the JSON’s first stored value is post-9-minibatch-updates and misses the true initial state). The visible spike at step ≈ 10 is **Adam’s first-update sign behaviour**: at a near-flat local minimum, $\hat{m}_1/\sqrt{\hat{v}_1} \approx \text{sign}(g_1)$ regardless of gradient magnitude, so the first 9 minibatch updates throw the operator off the minimum by an amount that scales with LR, not with gradient norm. Adam pulls it back to the blocked floor by step ≈ 100 and stays there for the remaining 900 steps. Log y compresses the resulting $\approx 10 \times$ dynamic range to keep both the transient and the convergence readable. Loss definition: $\sum_{ij} \left| x_{ij} - \left(T_\theta^\dagger \text{top}_K(T_\theta x) \right)_{ij} \right|^2$ summed over 65536 pixels of a 256×256 image at $K = 6554$ ($\rho = 0.10$), averaged over the 75-image validation split.

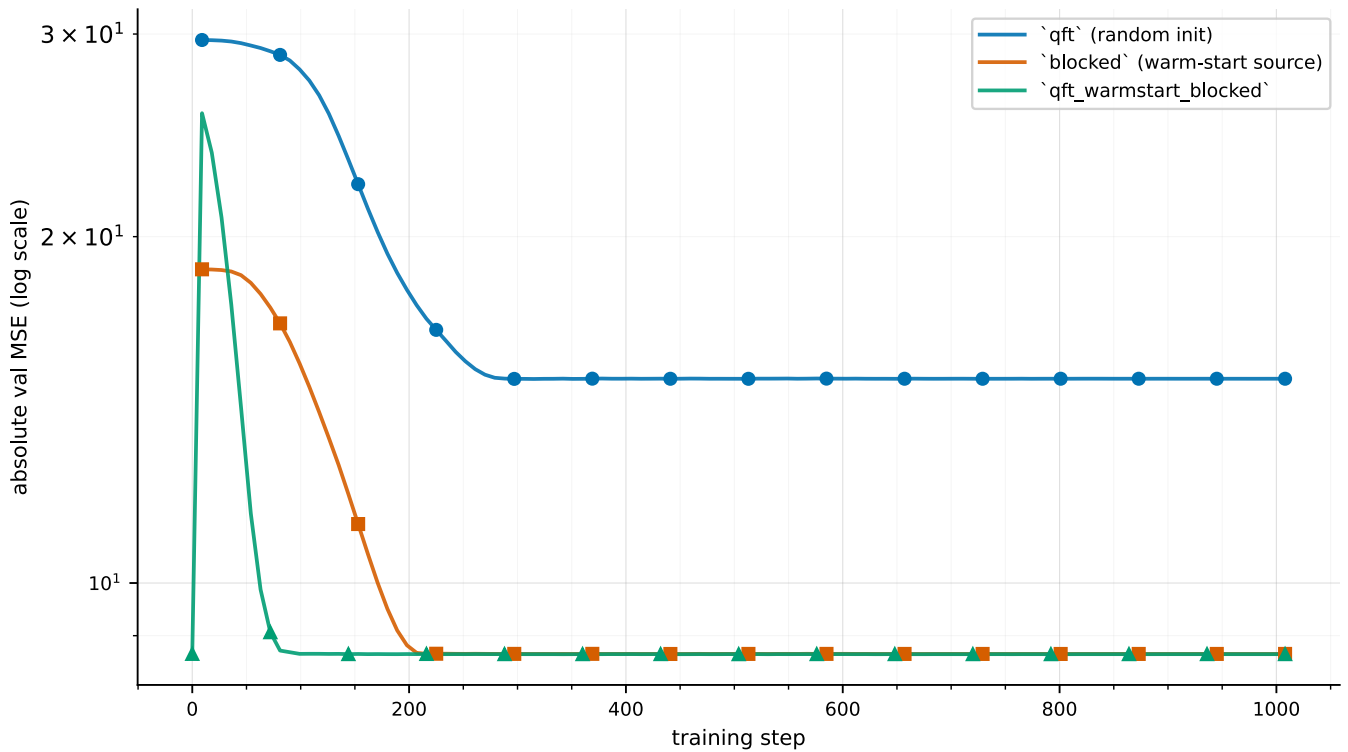


Figure 2: QuickDraw. Validation MSE per training step (log y). Blue: `qft` from random init. Orange: `blocked` headline run (warm-start source). Green: `qft_warmstart_blocked` — **step-0 marker plotted at the trained blocked floor** (bit-exact identity). The visible spike around step ≈ 10 is the same Adam first-update sign-behaviour described in the DIV2K caption; converges back to the blocked floor by step ≈ 200 and stays there. Loss definition: $\sum_{ij} \left| x_{ij} - \left(T_{\theta}^{\dagger} \text{top}_K(T_{\theta} x) \right)_{ij} \right|^2$ summed over 1024 pixels of a 32×32 image at $K = 102$ ($\rho = 0.10$), averaged over the 75-image validation split.

What this rules out. The headline `qft` < `blocked_8` gap is **not** a parametric-family limitation — the trained blocked operator lives inside $\text{QFT}(8, 8)$ and inside $\text{QFT}(5, 5)$, and Adam keeps it there once it starts there. The gap is therefore an optimisation-from-random-init phenomenon: the `qft` random init does not lie in the basin of the blocked optimum, and gradient descent does not cross between them in 1008 steps. (This experiment does not characterise **why** — it only confirms that the blocked optimum is achievable in the QFT family given the right starting point.)

Reproducibility. `experiments/qft_warmstart_blocked.py`

`--dataset {quickdraw, div2k_8q}`. The construction of the warm-start QFT lives in `pdf_t_benchmarks.bases.qft_warm_from_trained_blocked` and is asserted bit-exact against the loaded trained blocked basis at the start of every run.